BITS Pilani
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WORK INTEGRATED LEARNING PROGRAMMES

M.Tech. Automotive Electronics

for Working Professionals

M.Tech. Automotive Electronics is a four-semester Work Integrated Learning Programme designed for imparting knowledge and skills required to meet the evolving needs of the automotive industry in its quest for integrating technological advances in hardware, power electronics, cyber systems and machine intelligence. It is intended for engineering professionals working in the automotive OEM, auto components, electronics manufacturing and related industries desiring to broaden their knowledge horizons as they fulfill their career aspirations by contributing to this growing field.

The programme providesworking professionals with knowledge and expertise in multiple disciplines - Autotronics, automotive embedded systems and controls, automotive cyber systems, electric and hybrid architectures, autonomous vehicles, machine intelligence and Advanced Driver Assistance Systems (ADAS), – enabling them to design, build, test and sustain future automotive systems using digital technologies.

M.Tech. Automotive Electronics is a BITS Pilani Work Integrated Learning Programme (WILP). BITS Pilani Work Integrated Learning Programmes are UGC approved.

Who Should Apply?

The programme is designed for engineering professionals holding BE/ B.Tech or equivalent in domains such as Automotive, Mechanical, EEE, ECE, Instrumentation, Mechatronics and working in:

- Industries such as Automotive, Auto component, Design, Manufacturing, etc.
- Functional areas such as R&D, AnalysisMaintenance, Projects, component design etc.

What are the Highlights of the Programme?

- M.Tech. Automotive Electronics is a BITS Pilani Work Integrated Learning Programme (WILP). BITS Pilani Work Integrated Learning Programmes are UGC approved.
- The programme is of four semesters, with online classes conducted primarily during weekends and after-business hours. It can be pursued without career break.
- The programme has been designed for engineering professionals employed in industries such as Automotive, Auto component, Design, Manufacturing, etc., and working in functional areas such R&D, Analysis, Maintenance, Projects, component design, etc.
- The delivery methodology is a blend of classroom and experiential learning. Experiential learning consists of lab exercises, assignments, case studies and work integrated activities.
- The programme encourages the application of classroom knowledge to the solution of practical problems through its experiential learning component, preparing students to meet the challenges of problems of industry scale and complexity
- The programme offers "Virtual Simulation" using tools such as MATLAB/Simulink, Ricardo etc. and remote labs in automotive controls, cybersystems, electric vehicles, diagnostics and ADAS.
- Students are enrolled for four courses during each of the first three semesters. The final semester devoted to Dissertation/ Project Work enables the students to apply the concepts and skills acquired during the programme
- Students are assessed through a continuous evaluation process during the course of the semester, providing continuous feedback helping busy working professionals stay on course with the programme.
- Participants who successfully complete the programme will become members of an elite & global community of BITS Pilani Alumni.
- Option to submit fee using easy- EMI with 0% interest and 0 down payment.

What are the programme objectives?

M.Tech. in Automotive Electronics is designed for engineers working in the automotive OEM, auto components, electronics manufacturing and related industries who wish to prepare themselves for the future in the automotive electronics industry. The programme tends to equip professionals with multidisciplinary expertise in electronics, embedded systems, cyber systems, electric vehicles, autonomous vehicles and machine learning applications in the automotive industry.

What are the student learning outcomes?

Upon successful completion of the programme, learners will be able to:

- Analyze, select and implement suitable technologies for V2V and V2X communcations.
- Design and develop autotronic systems to control vehicle performance.
- Architect and design efficient system-on-chips for automotive applications.
- Develop and select diagnostics and control units for automotive systems.
- Architect electric and hybrid vehicle configurations and traction motors
- Understand automotive process frameworks supported by standards such as ISO

Learning methodology

Attend online lectures over weekends

Lectures are conducted live via online classes. These lectures can be attended via the internet using a computer from any location. These online classrooms offer similar levels of interactivity as regular classrooms at the BITS Pilani campus.

Classes for students admitted during the period Apr 2023 - July 2023 will begin in July/Aug 2023. The class schedule is announced within 1 week of completion of the admission process.

Online lectures are conducted usually over weekends for about 8 hours per week. Recorded content of a missed lecture may be accessed online.

Digital learning

Learners can access engaging learning material at their own pace which includes recorded lectures from BITS Pilani faculty members, course handouts and recorded lab content where applicable.

Continuous assessment

Continuous Assessment includes graded Assignments/Quizzes, Mid-semester exam, and Comprehensive Exam.

Experiential learning

The programme emphasizes on Experiential Learning that allows learners to apply concepts learned in the classroom to simulated and real work situations.

Control System Development

- Model-based generation from inside Simulink environment.
- Brake By Wire Test Rig – Using a single Master cylinder and fail-safe mode operation using conventional brake levers.
- Steer By Wire Test Rig – Uses Electronic Power Steering setup with “By Wire” input from Logitech G29 steering wheel.A2L file definitions.
- Configurable pin allocations, timers and interrupts.
- ECU calibration
- Control strategies for engines and transmissions

ADAS Test Rig

- Brake By Wire Test Rig – Using a single Master cylinder and fail-safe mode operation using conventional brake levers.
- Steer By Wire Test Rig – Uses Electronic Power Steering setup with “By Wire” input from Logitech G29 steering wheel.
- Driving cockpit interfaces with test rigs over CAN network and controlled by a slave VCU.

Electric Vehicle Demonstration Platform

- EV powertrain demonstration platform comprising of AC motor, Motor Controller, Li-Ion battery pack and gearbox.
- Control Algorithms of BLDC, Induction and PMSM Motors with Hardware in Loop (HIL).
- Battery Management System (BMS).

HIL Test Rig

- ★ Non-Real time HIL Test Rig.
- ★ Capable of running FMU compatible plant models.
- ★ Interfaces with standard vehicle controllers.

Capable of running on real-time HIL tests virtually on cloud servers using FMU compatible models.

Case studies and assignments

Carefully chosen real-world cases & assignments drive the problem-solving exercises

Dissertation/Project work

The Dissertation/Project work is the culmination of the programme in the fourth semester. It offers learners the opportunity to apply the knowledge they have gained during the programme to solve real-world like complex projects of value to their organizations.

VIRTUAL LAB

The virtual lab is a cloud-based simulation platform to design, develop and test solutions to engineering problems. It instils the requisite skills and ability to solve large scale complex problems encountered in the industry. The lab hosts a range of popular software tools to accurately model and simulate real engineering processes. The lab has live-support and is open 24x7x365.

LAB ARCHITECTURE

- The lab is organized around the following engineering domains: Controls and embedded systems, Electric and hybrid vehicles, Advanced Driver Assistance Systems (ADAS), Automotive cybersystems,
- The lab exercises are packaged as simulation-capsules that guide the students through a sequence of structured problems of increasing complexity.
- Participants may schedule their sessions and perform the experiments by accessing the lab remotely through the internet.
- On completing an experiment, the results may be downloaded and analyzed. The lab report may be uploaded in the learning management system for evaluation.

LEARNING OUTCOMES:

- Reinforce classroom learning through “hands-on” learning
- Build competency for solving industry-scale problems

Participants of M.Tech. Automotive Electronics can access one or more of the following software in their virtual labs:

FlexSim problem solved.

FRANC3D

WAVE

DFMA® BOOTHROYD DEWHURST

COMSOL

AUTOMATION STUDIO™

creo™

ELECTRIC & HYBRID VEHICLES VIRTUAL LAB

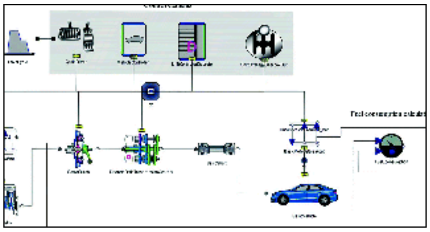
The Electric & hybrid vehicles virtual lab provides the environment for modeling the complete vehicle system in conventional, hybrid, electric and novel vehicle architectures using MATLAB/Simulink and the built-in powertrain library in Ricardo Ignite. It offers adjustable component fidelity for vehicle modeling – from initial concept to detailed powertrain integration.

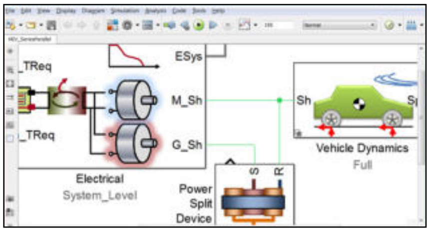
LIST OF EXPERIMENTS:

- Design of Electrical Vehicle (EV) model
- Introduction to power electronic converters design.
- Control Algorithms of BLDC, Induction and PMSM Motors

LEARNING OUTCOMES:

- Ability to model the entire vehicle system
- Model vehicle architecture





Software used:
Ricardo Ignite and
MATLAB & Simulink

MATLAB LAB

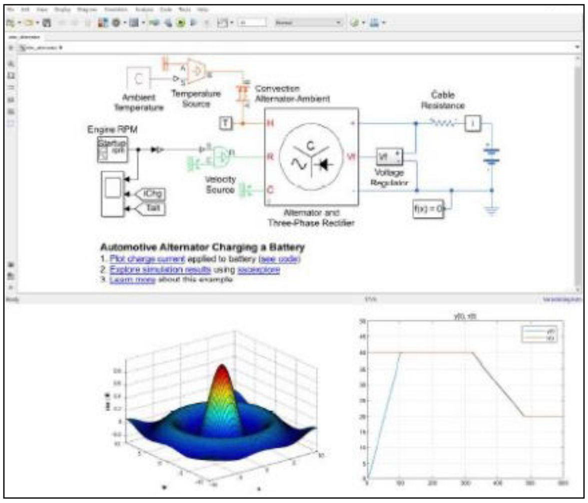
MATLAB (MATrix LABoratory) is a multi-paradigm numerical computing environment and proprietary programming language. MATLAB is used for computer programming, mathematical modelling such as controls, power electronics modeling etc. and Model In the Loop (MIL) simulations.

LIST OF EXPERIMENTS:

- Programming in MATLAB
- Mathematical modeling and simulation of dynamic systems
- PID controller design studies
- Model in the Loop (MIL) & Hardware In the Loop (HIL) testing using Simulink
- Modeling buck, boost and buckboost power converters

LEARNING OUTCOMES:

- ★ Develop plant models and control strategies for engineering applications and perform MIL testing
- ★ Develop programming knowledge and create replicas of real system using functional blocks



Software used:
MATLAB & Simulink

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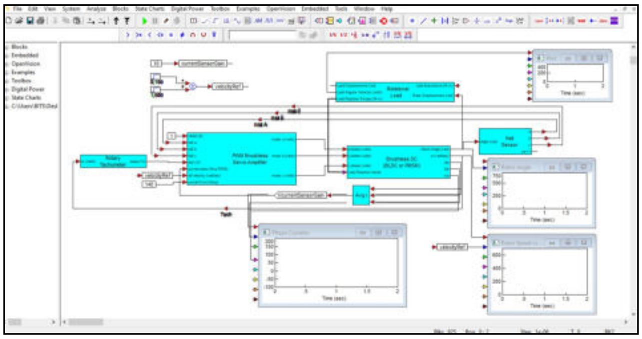
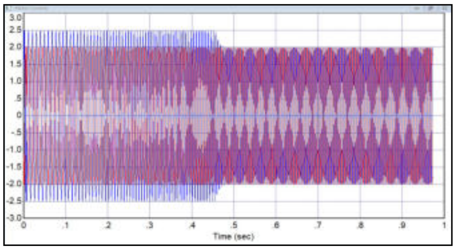
ALTAIR EMBED VIRTUAL LAB

Altair EMBED is a powerful model based development tool enabling users to design, analyze and simulate using block diagrams and state charts. The highly efficient diagram-to-code competence condenses the development time and increases the understanding of the vehicle subsystems. The quick development of virtual prototype of any dynamic system is an added feature.

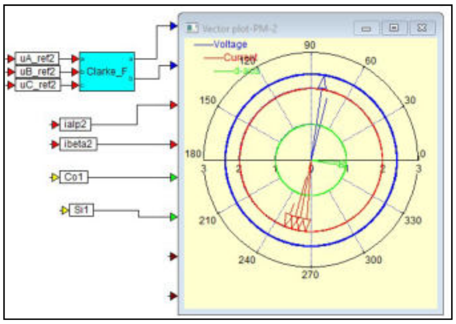


LIST OF EXPERIMENTS:



- ★ Design of Electrical Vehicle (EV) model
- ★ Introduction to power electronic converters design.
- ★ Control Algorithms of BLDC, Induction and PMSM Motors
- ★ Battery Management System (BMS).

LEARNING OUTCOMES:



- ★ Design a complete tool chain for development of Electric Vehicle (EV) covering software-in-the-loop, processor-in-loop as well as hardware-in-loop using model based development(MBD).
- ★ Modelling of interdependent operations of vehicle by compound blocks enabling final vehicle optimization.

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REMOTE LAB

The remote lab is an environment to perform controlled experiments from anywhere using IoT- enabled lab equipment and integrated remote access network. The lab is open 24x7x365. Remote Lab is yet another attempt by the WILP division of BITS Pilani ring the campus learning experience to working professionals complementing their work-life-learning balance.

LAB ARCHITECTURE:

- Participants schedule their session, access equipment remotely and perform the experiments by following detailed step by step instructions
- Participants can control the machines using equipment console while watching them through multiple cameras installed around
- Participants can pan, tilt or zoom to have a better view of the machines
- On completing an experiment, the results may be downloaded and analyzed. The lab report may be uploaded in the learning management system for evaluation.

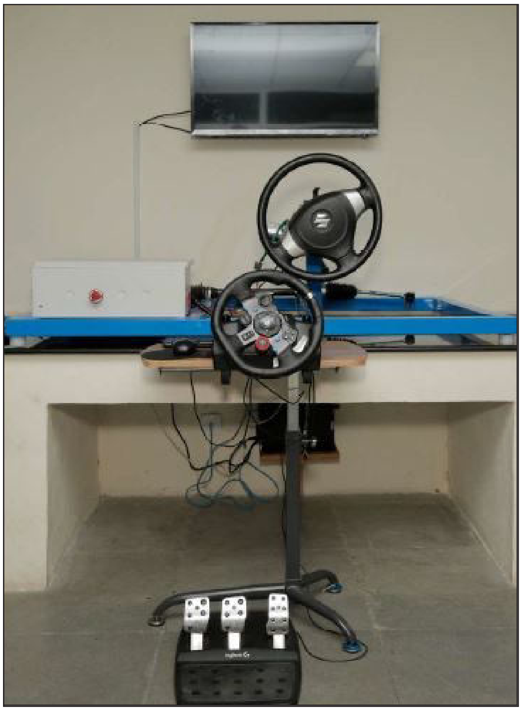
LEARNING OUTCOMES:

- Get real experience working with equipment
- Build competency for solving industry-scale problems

Participants of M.Tech. Automotive Electronics can access the following remote labs:

STEER-BY-WIRE TEST RIG

Steer-by-wire test rig comprises of a master and slave steering wheel setup that is controlled using MATLAB/Simulink environment via CAN communication. The actuators are controlled by Arduino microcontrollers in a closed-loop system based on sensor measurements.



LIST OF EXPERIMENTS:

- Develop a steering-control strategy for autonomous driving taking inputs from CAN messages
- Implement a PI controller to minimize the error and achieve target using a closed loop system

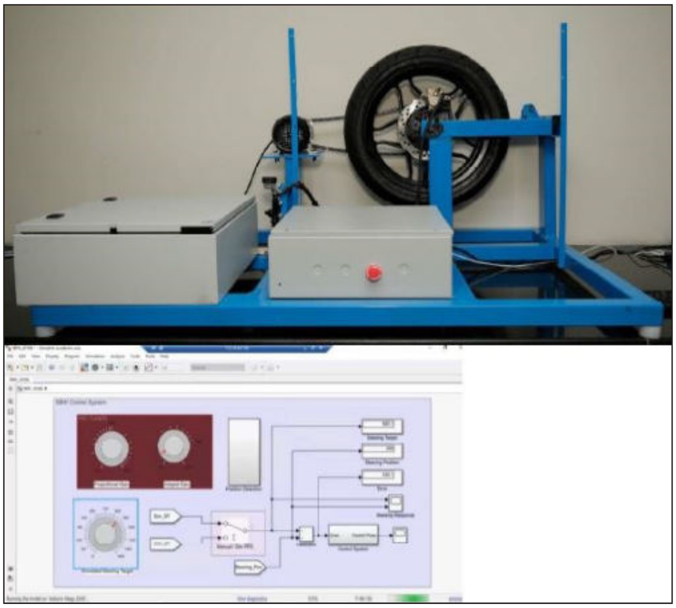
LEARNING OUTCOMES:

- Understand the behavior of PI based controllers in minimizing errors in a closed-loop system
- Develop/implement a control strategy to trace the required steering angle based on communicated inputs to the controller

BRAKE-BY-WIRE TEST RIG

Brake-by-wire test rig comprises of a single wheel controlled by a motor to simulate the braking system in hard braking conditions.

LIST OF EXPERIMENTS:

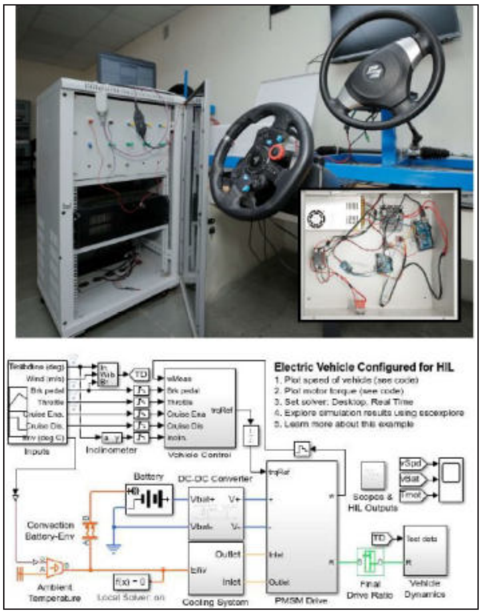


- ★ Develop a control strategy for applying brakes using desired slip ratio for different road-tire slip conditions
- ★ Implement a PI controller to minimize error and achieve target using a closed loop system

LEARNING OUTCOMES:

- ★ Understand the behavior of PI based controllers in minimizing errors in a closed-loop system
- ★ Develop/implement a control strategy for braking under various tire slip conditions

HIL (HARDWARE-IN-THE-LOOP) TEST RIG



HIL test rig consists of data acquisition for I/O signal communication for testing real-time hardware components integrated with the real-time plant model.

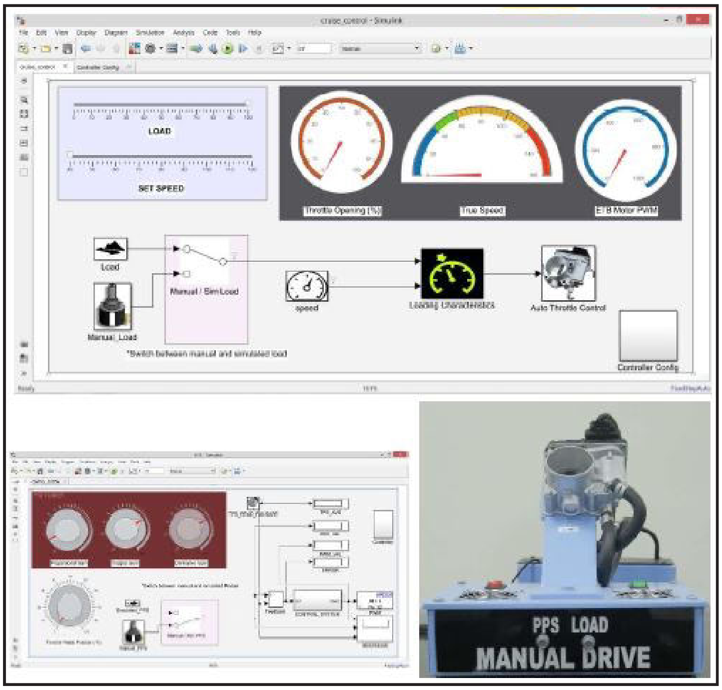
LIST OF EXPERIMENTS:

- ★ Tune the automotive engine controller using the mapped engine model
- ★ Design and develop a controller for EV & HEV
- ★ Test the hardware functionality of an electronic component in closed-loop

LEARNING OUTCOMES:

- ★ Understand the process of hardware testing as an important phase of vehicle electronics and software systems development
- ★ Develop various test scenarios within operating range and calibrate for optimum performance

ELECTRONIC THROTTLE BODY TEST RIG



The ECU-based electronic throttle body is used to explore various control strategies to obtain the desired throttle valve position and valve opening rate in a modern electronic throttle body. The control algorithm is modeled in MATLAB/Simulink.

LIST OF EXPERIMENTS:

- ★ Modeling control strategies such as cruise and fail-safe
- ★ Manual and automatic PID tuning of throttle position

LEARNING OUTCOMES:

- ★ Detect instability, overshoot and other control parameters while tuning the controller
- ★ Design control strategies in MATLAB/Simulink and implement different configurations in the hardware

AUTOMATIC GEAR BOX TEST RIG

This test rig focuses on building a complex strategy to operate a gearbox under many conditions such as steep gradients, excess speed and overload.

LIST OF EXPERIMENTS:

- ★ Develop control strategies for automatic gear box
- ★ Develop control strategy for uphill mode and several failsafe modes in MATLAB/Simulink
- ★ Implement the optimum control strategy in the hardware and perform HIL test

LEARNING OUTCOMES:

- Understand the control of automatic transmission
- Understanding the control algorithm and tweaking the control parameters
- Implementation of failsafe modes

AUTOMOTIVE CYBER SYSTEMS (ACS) LAB

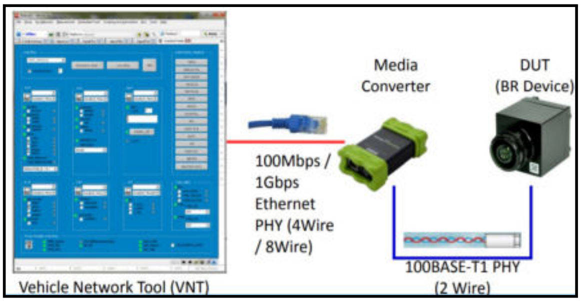
The ACS lab aims at providing a complete knowledge and first-hand experience of automobile to the students of Automotive Engineering & Electronics. The lab uses both virtual tools and hardware. Students use them to understand the CAN (Controller Area Network), CAN FD, LIN (Local Interconnect Network), Ethernet networks, ECU Flashing & CAN Fuzz (Software Penetration & Security Testing), Automotive Ethernet & DoIP, Automotive Ethernet Switches.

LIST OF EXPERIMENTS:

Automotive Communication Systems, Automotive Networking and Automotive Diagnostics.

IN VEHICLE NETWORKING (CAN, LIN & CAN FD)

- Actuating a motor using CAN protocol
- Basics of CAN, LIN & CAN FD – exercises
- CAN Protocols (UDS, KWP2000, J1939, J1979, CCP, XCP)
- Different databases & type of file formats
- Simulation
- Data Monitoring
- Signal plots
- Scripting & automation
- Data logging and analysis

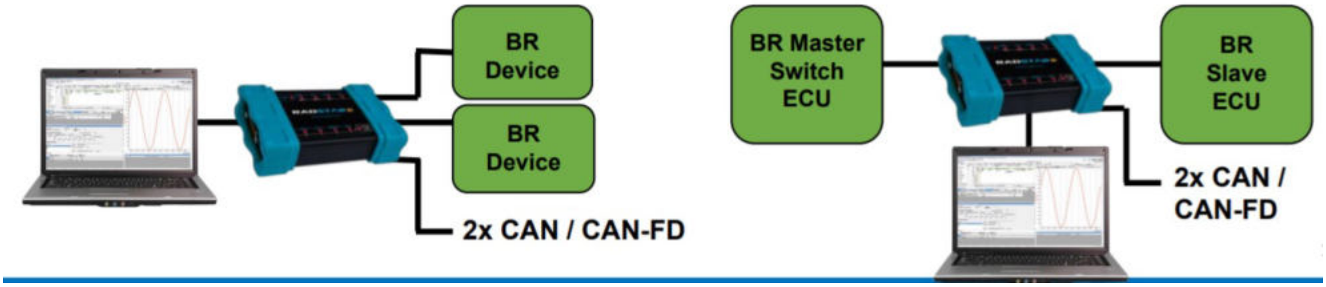


AUTOMOTIVE ETHERNET & DoIP

- ★ Automotive Ethernet- Overview, concepts, how it's different than other networks (CAN, CAN FD, LIN, Flex ray)
- ★ Create your own AE network
- ★ Know about UDP, TCP, ICMP, IGMP, ARP protocols
- ★ Create ARP, TCP, Server, Client example
- ★ DoIP- protocol details
- ★ DoIP- setup and Edge node, ECU & test configuration
- ★ Run DoIP setup, test DoIP services

AUTOMOTIVE ETHERNET SWITCHES

- ★ Media conversion
- ★ Frame Mirroring and VLAN tagging for advanced debug and Monitoring
- ★ Gateway applications

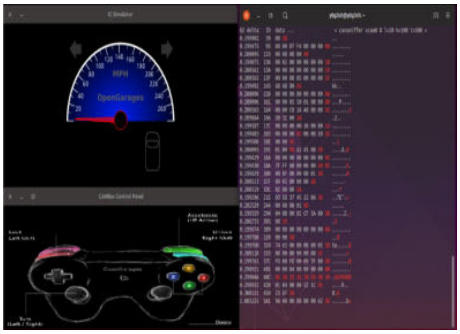


BUILD YOUR OWN CAN SOFTWARE TOOL

- ★ Use of neoVI Dll's to create own CAN software tool
- ★ Cross platform API's- Python, VC++, Excel, VB etc.
- ★ Custom software application creation

ECU FLASHING & CAN FUZZ (SOFTWARE PENETRATION & SECURITY TESTING):

- ★ Concepts of ECU Flashing Flash Sequences
- ★ Flash Configuration
- ★ Flashing process & analysis
- ★ Concepts of CAN Fuzzing
- ★ CAN Fuzz logic and Test creation
- ★ CAN Fuzz on CAN ID, DLC, Data etc
- ★ Fuzz type- Random, Counter, Fix



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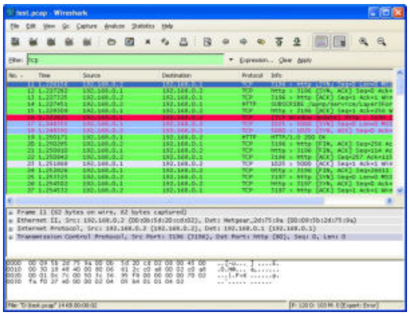
V-CAN: Sniff CAN communication, Identify CAN ID for various functions, hack the virtual system.

WIRESHARK: Analyse the tapped CAN Dump using Wireshark tool to understand the vulnerabilities and analyse communication protocol.

LEARNING OUTCOMES:



- ★ Tap into CAN Communication networks
- ★ Understand CAN, LIN, Ethernet Communication protocols
- ★ Identify CAN IDs of a given function/ system
- ★ Penetrate CAN communication
- ★ Execute denial of service/ malfunction
- ★ Understand how to flash an ECU and CAN Fuzzing
- ★ Creating own CAN software tool, AE network

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ADVANCED DRIVER ASSISTANCE SYSTEMS (ADAS) LAB

The objective of the lab is to familiarize the students with the latest technology supporting driver assistance systems.

LIST OF EXPERIMENTS:

- Distance measurement and Audio and Visual alert.
- Object detection and classification using CNN
- Virtual Experiments based on computer vision technique
- Implementation of Hough transform based lane detection using python-opencv libraries.
- Traffic Sign Classification using deep neural networks.
- Python based implementation of Kalman filters.

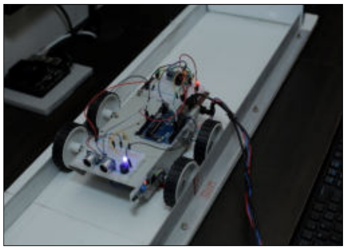
LIST OF EXPERIMENTS FOR RISD:

- Python/Matlab imp lementation of discrete Fourier transform.
- Segmentation of ground plane in Lidar point cloud using RANSAC algorithm.
- Semantic segmentation of urban scenes using U-Net.

ADAS LAB COVERS

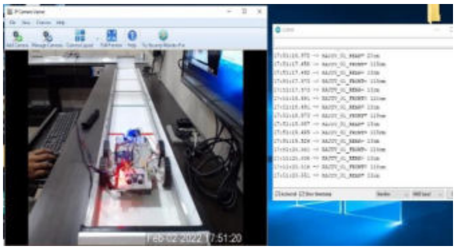
BASICS OF ELECTRONICS EXPERIMENT:

- Distance measurement and Audio and Visual alert.



Boat equipped with LED, Buzzer and Ultrasonic sensor with Arduino board where students can write there program

Interface created for Motion control of boat remotely.



Output: distance and different Audio Visual alerts

EXPERIMENTS BASED ON CNN:

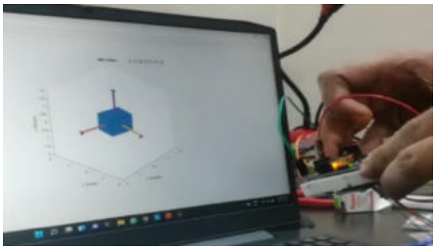
- ★ Object detection and classification using CNN (Convolutional neural networks)

VIRTUAL EXPERIMENTS BASED ON COMPUTER VISION TECHNIQUE

- ★ Tools and learning outcomes
  - Python programing
  - Image processing
  - Computer vision Techniques

EXPERIMENTS BASED ON SENSOR FUSION

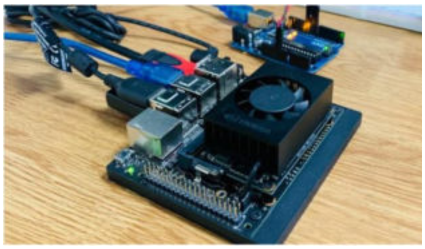
- ★ Sensor fusion using IMU and GPS sensors for self localization of ego vehicle.



DESIGN AND DEVELOPMENT OF ADAS USING RADAR AND LIDAR SENSORS.

Computing Hardware we use

- ★ Nvidia Jetson Xavier Development kit.
- ★ Nvidia Jetson Xavier Nano



Hardware Capability

- ★ A powerful AI computer
- ★ It is designed specifically for autonomous machines

It can handle

- ★ Visual odometry
- ★ Sensor fusion
- ★ Localization and mapping,
- ★ Obstacle detection and path planning algorithms

ELECTRIC AND HYBRID VEHICLE REMOTE LAB

The Electric and hybrid vehicle lab provides the sophisticated technical environment for understanding and modelling the complete control algorithm for different motors mostly used for xEV's design using Altair embed, also to look at complete understanding for engineers could develop BMS algorithms by performing simulation in loop (SIL) to hardware in loop (HIL).

LIST OF EXPERIMENTS:

- Control Algorithms of BLDC, Induction and PMSM Motors with Hardware in Loop (HIL).
- Battery Management System (BMS).

LEARNING OUTCOMES:

- Understanding of complete control algorithms for different motors in Electric Vehicle (EV) covering voltage & current controller, sensor FOC and Sensorless FOC using model based development (MBD).
- Validating and implementation of SIL to HIL of complete battery performance involves several measurements.

Mode of Examinations

Examinations Mode Options applicable for students admitted in Batch starting in July/Aug 2023.

Semester 1, 2 and 3 have Mid-Semester Examinations and Comprehensive Examinations for each of course. These examinations are mostly scheduled on Friday, Saturday or Sunday. During these semesters, in addition to the Mid-Semester and Comprehensive examinations, there will also be Quizzes/Assignments conducted online on the Learning Management System (LMS) as per the course plan in which the students need to participate. In Semester 4 (Final Semester), the student will be doing a Dissertation/Project Work as per the Institution's guidelines.

Two Options on Mode of Examinations:

It is institute's endeavour to offer two options on mode of exams at the time of the registration for each semester of the programme. However, please note that Option 2 as explained below is offered purely on the discretion of the institute. Availability of Option 2 is not assured and is subject to institute's own assessment of the feasibility of providing it. Students will need to choose one from the available option/s provided by the institute at the time of registering for every semester. The two mode of exam options are explained below:

Option 1 - Examinations at Designated Examination Centres:

Students choosing this option will need to appear in person for taking the examinations at the institution's designated examination centres as per the examination schedule, instructions, rules and guidelines announced before every examination. These designated examination centres are at the following locations: **Bangalore, Chennai, Delhi NCR, Hyderabad, Pune, Mumbai, Kolkata, Goa, and Pilani**. In addition to these locations, Institution also has a designated examination centre in **Dubai**.

Option 2 - Online Examinations:

Students choosing this option can take the examinations online from any location e.g. office or home. **However the Institute at its discretion may choose not to offer this option and in such a case students will need to take the examinations as per Option 1 that entails appearing for the examination at any one of the designated examination centres.** Also, to take an Online Examination, the student must possess a Laptop or Desktop system with Two Web Cams (One Web Cam for the student's frontal face view and a Second Web Cam for the student's and Laptop or Desktop system's full side view during the exam), a smart phone and good internet connectivity.

Please **click here** to refer to the complete details about mandatory IT and Non-IT Infrastructure requirements for taking the online examinations. A student should choose this option only if the student is confident to arrange the required mandatory IT Infrastructure and Non-IT Infrastructure to take the examinations under this mode. Students opting for online examinations would require to log in to the institution's online examination platform as per the announce examination schedule and take the online examinations in compliance with the institution's defined instructions, guidelines, and rules which are announced before all examinations.

Important:

While it will be institute's endeavour to offer both the above options for students to choose from at the time of registering for any semester, however availability of the online mode option may not always be feasible or is assured and Institute at its own discretion could choose to offer only Option 1 as mentioned above. Also note that The Institute regularly takes actions to optimize its examination system and hence the mandatory IT and Non-IT Infrastructure requirements, instructions, guidelines, and rules associated with both the above mentioned examination modes may change anytime at the Institute's discretion. All students will need to 100% comply with any such changed specifications announced by the Institute.

What is the Eligibility Criteria?

Minimum eligibility to apply: Applicants must be employed professionals working in Engineering organizations, and holding B.E./ B.Tech. or equivalent in domains such as Automotive/ EEE /ECE/ Instrumentation/ Mechatronics/ Mechanical with at least 60% aggregate marks. A minimum of 1 year of work experience in relevant domains after the completion of the degree is required to apply.

The programme is designed for engineering professionals holding BE/ B.Tech or equivalent in domains such as Automotive, Mechanical, EEE, ECE, Instrumentation, Mechatronics and working in:

- Industries such as Automotive, Auto component, Design, Manufacturing, etc.
- Functional areas such as R&D, Analysis, Maintenance, Projects, component design etc.

Fee Structure

The following fees schedule is applicable for candidates seeking new admission during the academic year 2023-24:

Application Fees
(one time)
₹1,500

Admission Fees
(one time)
₹16,500

Semester Fees
(per semester)
₹63,500

EMI with 0% interest and 0 down payment

Instant EMI option with 0% interest and 0 down payment is now available that allows you to pay programme fee in an easy and convenient way.

- Instant online approval in seconds
- No Credit Cards/ CIBIL score required
- Easy & Secure online process using Aadhaar and PAN number
- Anyone with a Salary Account with Netbanking can apply
- Option to submit fee using easy- EMI with 0% interest and 0 down payment

All the above fees are non-refundable.

Important: For every course in the program institute will recommend textbooks, students would need to procure these textbooks on their own.

Programme Curriculum

Participants need to take at least 12 courses towards coursework and complete one Project/ Dissertation. The coursework requirement for the programme would consist of a set of core courses and electives. Core courses are compulsory for all participants, while electives can be chosen based on individual learning preferences.

First Semester

- Automotive Vehicles
- Autotronics
- Embedded System Design
- Automotive Communication Systems

Second Semester

- Advanced Driver Assistance Systems
- Automotive Control Systems
- Automotive Networking
- Elective 1

Third Semester

- Elective 2
- Elective 3
- Elective 4
- Elective 5

Fourth Semester

- Dissertation

Electives

- Automotive Systems Engineering
- Electric & Hybrid Vehicles
- Automotive Security
- Automotive Diagnostics & Interfaces
- Product Design
- Safety Critical Advanced Automotive Systems
- Robust and Intelligent Systems Design

Choice of Electives is made available to enrolled students at the beginning of each semester. A limited selection of Electives will be offered at the discretion of the Institute.

Detailed Course Curriculum

Automotive Vehicles

- Internal combustion engines; vehicle performance; analysis and design of vehicle components. Experimental or theoretical investigation of problems selected from the field of automotive vehicles.

Autotronics

- Fundamentals of automotive EMC, control concepts, control design with the help of sensors and signal conditioning. Understanding of autotronics and vehicle intelligence, sensor technologies, intelligent systems and mechatronic modelling. Introduction, electricity and electronic fundamentals, sensors, sensor types, signal conditioning, system modelling, dynamic response of systems, feedback/closed loop controllers, electronic fuel control systems, actuators: fuel injectors, exhaust gas recirculation, motors and ignition systems, hydraulics.

Automotive Systems Engineering

- Automotive systems development and testing, compatibility issues, performance prediction, design requirements and engineering metrics, systems engineering process, life cycle standards and management, concurrent engineering, systems analysis applications, and advanced model based development.

Embedded System Design

- Introduction to embedded systems; embedded architectures: Architectures and programming of microcontrollers and DSPs. Embedded applications and technologies; power issues in system design; introduction to software and hardware co-design.

Electric and Hybrid vehicles

- Electric motors, drives, control, batteries, architectures, energy storage, recovery, and management, characteristics of autonomous vehicles, modelling, simulation, analysis and comparison of relations among multiple parameters for electric, hybrid and autonomous vehicles, insights into regulations and norms with respect to electric, hybrid and autonomous vehicles, hybrid vehicle propulsion systems, sustainable automotive power technology.

Automotive Communication Systems

- Introduction to communication engineering; automotive communication systems: basic, current and future generation automotive communication protocols and telematics, advanced communication, intersystem communication and multiplex systems, wireless and photonics systems engineering, communications and networking, signal propagation in a mobile environment, modulation, coding, equalization, multiple access techniques, spread spectrum systems, second and third generation systems, UMTS, IMT-2000; Intra Vehicular Buses - CAN, TTCAN, FTTCAN, RT and FT Ethernet, TTP/A, TTP/C, Flexray, LIN, MOST; Clock Synchronization and Diagnostic Services in Intra Vehicular Buses.

Automotive Networking

- ★ Overview of TCP/IP systems, Disturbed and Networked Embedded systems, Embedded Internet, Real Time Networks and Fault Tolerant Networks – Issues and Design, Intelligent Transport Systems and IoT for Automotive Systems; Fault and Error Containment, Intra and Interworking in Vehicular Systems: Intra Vehicular Buses - an overview, Time Triggered and Event Triggered Networks in Intra Vehicular Systems, Automotive Network Domains – Power Train, Chassis, Body Domains – Network Characteristics and Domain Requirements, V2I/V2R, V2V , VANETS - MANETS vs VANETS, Safety Applications vs Comfort Applications of VANETS, Network Architecture, Protocols, Network Stack, MAC protocols, IEEE Wave and DSRC, Routing Protocols, Network Security – Attacks and Solutions. Emerging and advanced automotive networks – Aerial Networks; Interconnection between various networks in ITS – Interconnection between Intra and Inter Vehicular Systems, Network Models in Automotive Systems – Publisher Subscriber Model, Producer Consumer Models, Device Interoperability Issues in Interconnected Vehicles, Middleware in Automotive Systems, Network Management Function, Objects and Device Management - AutoSar and Networked OS; Protocol-independent design methodology for distributed real-time networks in vehicles – Volcano.

Automotive Control Systems

- ★ Introduction to vehicle electronics, semiconductor diodes, FETs, rectifiers, small signal amplifiers, circuit models, automotive applications and case studies, automotive micro controllers, auto sensors and actuators, vehicle electronics, feedback control, control strategy, analog and digital controllers, expert systems and neural networks, advanced topics in EMC, vehicle communication networks, automotive control system design, transmission and powertrain, brake, traction, suspension, active safety and supplementary restraint systems, intelligent vehicle systems and ADAS.

Advanced Driver Assistance Systems

- ★ Automotive safety systems, assist and autonomous systems, automotive sensors and actuators for ADAS (stereo and mono cameras ultrasonic sensors, LIDAR, RADAR), fundamentals of machine vision, data fusion for ADAS, mechatronics for ADAS, human – machine interface for ADAS, telematics and infotainment, ADAS system, legal and ethical aspects of ADAS, real time systems and development, advanced driver assistance systems, advanced computer systems, automated driving applications and systems.

Automotive Security

- ★ Security concepts, security attacks and risks, architectures, policy management, mechanisms, understanding the risks and advantages of vehicle to internet (V2I), vehicle to vehicle (V2V), vehicle to IoT (V2IoT) connectivity, issues concerning the security of intelligent transport systems that communicate with the vehicle, telematics, cryptography, security standards, security system interoperation and case studies of the automotive security systems and connectivity technologies, automotive cyber security and autonomous vehicles, connected vehicle driver responsibility, issues around liabilities related to automotive cyber security incidents.

Automotive Diagnostics and Interfaces

- ★ Sensors used in today's vehicles, such as temperature, pressure, position, distance, velocity, torque and flow; Designing and building analogue interfaces, regulation and control problem with reference to power electronic converters; converter models for feedback: basic converter dynamics, fast switching, piece-wise linear models, discrete-time models, On board diagnostics II (OBD II); Voltage mode and current mode controls for DC-DC converters, comparator based control for rectifier systems, proportional and proportional-integral control applications; Control design based on linearization: transfer functions, compensation and filtering, compensated feedback control systems; Hysteresis control basics, and application to DC-DC converters and inverters; Automotive diagnostics, electronic interfaces, sensors and interfacing, introduction to microsystems packaging, microcomputer control systems, reliability, diagnostics, and testing of vehicles.

Product Design

- ★ Introduction to creative design; user research and requirements analysis, product specifications, Computer Aided Design; standardization, variety reduction, preferred numbers and other techniques; modular design; design economics, cost analysis, cost reduction and value analysis techniques, design for production; human factors in design: anthropometric, ergonomic, psychological, physiological considerations in design decision making; legal factors, engineering ethics and society.

Safety Critical Advanced Automotive Systems

- ★ Functional safety, safety in electrical engineering, architecture / design practices for safety critical systems, ISO 26262: Road vehicles – functional safety, IEC 1508 standards; Methodology of certification and qualification for IEC 1508, modelling real time systems (UML-RT, and the tools), reliable, common system bus – VME, ASCB, safeBus, multiBus II etc. Real time and safety standard and certifications, FPGA and ASIC based design, low-power techniques in RT embedded systems on-chip networking; Hardware software partitioning and scheduling, co-simulation, synthesis and verifications, architecture mapping, HW-SW interfaces.

Dissertation

- ★ A student registered in this course must take a topic in an area of professional interest drawn from the on-the-job work requirement, which is simultaneously of direct relevance to the degree pursued by the student as well as to the employing/ collaborating organization of the student and submit a comprehensive report at the end of the semester working under the overall supervision and guidance of a professional expert who will be deemed as the supervisor for evaluation of all components of the dissertation.

  Normally the Mentor of the student would be the Dissertation supervisor and in case Mentor is not approved as the supervisor, Mentor may play the role of additional supervisor. The final grades for dissertation are Non-letter grades namely Excellent, Good, Fair and Poor, which do not go into CGPA computation.

Robust and Intelligent Systems Design

- ★ Principles of fault tolerant systems, redundancy, parallel and shared resources, spatial systems, configurations, design aspects, intelligent transport systems, neural networks and fuzzy logic, reconfigurable hardware system design, energy aware computing systems.

How to apply

- ★ Create your login at the Application Center by entering your unique Email id and create a password of your choice.

- ★ Once your login has been created, you can anytime access the online Application Center using your email ID and password. Once you have logged in, you will see a screen showing 4 essential steps to be completed to apply for the programme of your choice.

- ★ Begin by clicking on Step 1 - ‘Fill/ Edit and Submit Application Form’. This will enable you to select the programme of your choice. After you have chosen your programme, you will be asked to fill your details in an online form. You must fill all details and press ‘Submit’ button given at the bottom of the form.

- ★ Take the next step by clicking on Step 2 - 'Download Application PDF Copy’. This will download a pdf copy of the application form on your computer.

- ★ Now, click on Step 3 - 'Pay Application Fee’ to pay INR 1,500/- using Net banking/ Debit Card/ Credit Card.

- ★ Take a printout of the downloaded Application Form and note down the Application Form Number that appear on the top-right corner of the first page. This Application Form Number should be referred in all future correspondence with BITS Pilani.

- ★ In the printout of the downloaded Application Form, you will notice on page no. 3 a section called the Employer Consent Form. Complete the Employer Consent Form. This form needs to be signed and stamped by your organisation’s HR or any other authorised signatory of the company.

Important: In view of work-from-home policies mandated by many organisations, a few candidates may not be able to get the physical forms signed by their HR/ other authorised organisational representative. Such candidates may instead request an email approval to be sent to their official email ID by the HR using the format available through this link.

- ★ Further on page no. 4 of the printed Application Form is a section called the Mentor Consent Form. The Mentor Consent Form needs to be signed by the Mentor.

Important: In view of work-from-home policies mandated by many organisations, a few candidates may not be able to get the physical forms signed by their Mentor. Such candidates may instead request an email approval to be sent to their official email ID by the Mentor using the format available through this link.

Who is a mentor:

Candidates applying to Work Integrated Learning Programmes must choose a Mentor, who will monitor the academic progress of the candidate, and act as an advisor & coach for successful completion of the programme.v

How to apply

Candidates should ideally choose the immediate supervisor or another senior person from the same organisation. In case a suitable mentor is not available in the same organisation, a candidate could approach a senior person in another organisation who has the required qualifications. Wherever the proposed Mentor is not from the same employing organization as that of the candidate, a supporting document giving justification for the same should be provided by the candidate's employer.

Candidates applying to B.Tech. programmes should choose a Mentor who is an employed professional with B.E./ B.S./ B.Tech./ M.Sc./ A.M.I.E./ Integrated First Degree of BITS or equivalent. Candidates applying to M.Tech., M.Sc., MBA, M.Phil programme should choose a Mentor who is an employed professional with:

- B.E./ M.Sc./ M.B.A./ M.C.A./ M.B.B.S. etc. and with a minimum of five years of relevant work experience

OR

- M.E./ M.S./ M.Tech./ M.Phil./ M.D./ Higher Degree of BITS or equivalent

★ Further on page no. 5 of the downloaded Application Form, is a Checklist of Enclosures/ Attachments.

- Make photocopies of the documents mentioned in this Checklist
- Applicants are required to self-attest all academic mark sheets and certificates

★ Finally, click on Step 4 - 'Upload & Submit All Required Documents'. This will allow you to upload one-by-one the printed Application Form, Mentor Consent Form, Employer Consent Form, and all mandatory supporting documents and complete the application process. Acceptable file formats for uploading these documents are .DOC, .DOCX, .PDF, .ZIP and .JPEG.

★ Upon receipt of your Application Form and all other enclosures, the Admissions Cell will scrutinise them for completeness, accuracy and eligibility.

★ Admission Cell will intimate selected candidates by email within two weeks of submission of application with all supporting documents. The selection status can also be checked by logging in to the Online Application Centre.

UGC Approval

BITS Pilani is an Institution of Eminence under UGC (Institution of Eminence Deemed to be Universities) Regulations, 2017. The Work Integrated Learning Programmes (WILP) of BITS Pilani constitutes a unique set of educational offerings for working professionals. WILP are an extension of programmes offered at the BITSPilani Campuses and are comparable to our regular programmes both in terms of unit/credit requirements as well as academic rigour. In addition, it capitalises and further builds on practical experience of students through high degree of integration, which results not only in upgradation of knowledge, but also in up skilling, and productivity increase. The programme may lead to award of degree, diploma, and certificate in science, technology/engineering, management, and humanities and social sciences.

On the recommendation of the Empowered Expert Committee, UGC in its 548th Meeting held on 09.09.20 has approved the continued offering of BITS Pilani's Work Integrated Learning programmes.

BITS Pilani
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WORK INTEGRATED LEARNING PROGRAMMES

Let's start a conversation to ignite the change you desire

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